

Divergent orthoprojections in a strictly convex Banach space

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Source question

The source paper is Catalin Badea and Yuri I. Lyubich, *Geometric, spectral and asymptotic properties of averaged products of projections in Banach spaces*, arXiv:1006.2052. On page 6, after giving a divergent example in ℓ_∞^2 , the authors write:

Then

$$(P_1 P_2)^n = \begin{pmatrix} 0 & (-1)^{n+1} \\ 0 & (-1)^n \end{pmatrix},$$

so the iterates $(P_1 P_2)^n$ are divergent.

The space in our example is not strictly convex. An open question is about existence of an example of divergence in a strictly convex space. For the affirmative answer the space must be infinite-dimensional since every finite-dimensional strictly convex space is uniformly convex.

For any Banach space X and its closed subspace M , we denote by $P_M(x)$, $x \in X$, the set of points in M whose distance to x is minimal. If X is reflexive then the set $P_M(x)$ is not empty for every x . If, in addition, X is strictly convex, then $P_M(x)$ is a singleton. In this situation $P_M(x)$ can be considered

In transcription, the question asks whether there is an example of divergence of iterates of a product of orthoprojections in a strictly convex Banach space. Here an orthoprojection means a non-zero norm-one linear projection.

Proof Intuition

The two-dimensional ℓ_∞^2 example has product eigenvalue -1 . Strict convexity forbids such an exact finite-dimensional example, because finite-dimensional strictly convex spaces are uniformly convex and the source paper's convergence theorem applies. But it does not forbid a sequence of two-dimensional blocks whose product eigenvalues are $-\rho_n^2$ with $\rho_n \uparrow 1$.

We put these blocks into an ℓ_∞ -sum. To make the sum strictly convex without destroying contractivity of the block projections, we add a weighted ℓ_2 term built from the block norms:

$$\|x\| = \left(\|x\|_\infty^2 + \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_n^2 \right)^{1/2}.$$

The block projections remain contractions for both terms. The vector consisting of the distinguished block eigenvectors then witnesses divergence: for every m , some far-out block still has ρ_n^{4m} almost equal to 1, so the even and odd powers of the product remain separated.

Lemma (two-dimensional blocks). *For every $0 < \rho < 1$ there is a real two-dimensional strictly convex norm $\|\cdot\|_\rho$ on $\mathbb{R}^2$ and two norm-one projections P_ρ, Q_ρ such that $P_\rho Q_\rho$ has eigenvalue $-\rho^2$.*

Proof. Let

$$u_\rho = (1, -\rho), \quad v_\rho = (\rho, 1)$$

in the coordinate plane with coordinate functionals e_1^*, e_2^* . We first choose a centrally symmetric strictly convex body K_ρ such that

$$\text{conv}\{\pm u_\rho, \pm v_\rho\} \subset K_\rho \subset [-1, 1]^2$$

and $u_\rho, v_\rho \in \partial K_\rho$.

Such a body is obtained by starting with the parallelogram $\text{conv}\{\pm u_\rho, \pm v_\rho\}$. Each open side of this parallelogram lies in the interior of the square $[-1, 1]^2$, because $0 < \rho < 1$. Replace each side by a small outward circular arc lying still inside the square, and do so centrally symmetrically. The resulting closed convex body has no boundary line segment, is centrally symmetric, remains inside the square, and keeps $\pm u_\rho, \pm v_\rho$ on its boundary. Its Minkowski functional is the desired strictly convex norm.

Since $K_\rho \subset [-1, 1]^2$, the coordinate functionals have norm at most 1 for $\|\cdot\|_\rho$. Since $u_\rho, v_\rho \in \partial K_\rho$, we have $\|u_\rho\|_\rho = \|v_\rho\|_\rho = 1$. Define

$$P_\rho x = e_1^*(x)u_\rho, \quad Q_\rho x = e_2^*(x)v_\rho.$$

Then $P_\rho^2 = P_\rho$ and $Q_\rho^2 = Q_\rho$, and

$$\|P_\rho x\|_\rho = |e_1^*(x)| \leq \|x\|_\rho, \quad \|Q_\rho x\|_\rho = |e_2^*(x)| \leq \|x\|_\rho.$$

Thus P_ρ and Q_ρ are norm-one projections. Finally,

$$Q_\rho u_\rho = -\rho v_\rho, \quad P_\rho Q_\rho u_\rho = -\rho^2 u_\rho,$$

so $-\rho^2$ is an eigenvalue of $P_\rho Q_\rho$. □

Counterexample. *There is a strictly convex real Banach space X and two orthoprojections $P, Q \in \mathcal{B}(X)$ such that the iterates $(PQ)^m$ do not converge strongly.*

Proof. Choose $0 < \rho_n < 1$ with $\rho_n \uparrow 1$. Let $E_n = (\mathbb{R}^2, \|\cdot\|_{\rho_n})$, and let $P_n = P_{\rho_n}$, $Q_n = Q_{\rho_n}$, and $u_n = u_{\rho_n}$ be as in the lemma. Let

$$Y = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell_\infty}$$

with $\|x\|_\infty = \sup_n \|x_n\|_{\rho_n}$. Define a new norm on Y by

$$\|x\|_X = \left(\|x\|_\infty^2 + \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{\rho_n}^2 \right)^{1/2}.$$

This norm is equivalent to $\|\cdot\|_\infty$, hence $X = (Y, \|\cdot\|_X)$ is a Banach space.

The weighted ℓ_2 term

$$G(x) = \left(\sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{\rho_n}^2 \right)^{1/2}$$

is strictly convex, because every block norm $\|\cdot\|_{\rho_n}$ is strictly convex and every block has positive weight. If equality held in the convexity inequality for $\|\cdot\|_X$, equality would also hold for G , forcing equality in every non-zero block and hence $x = y$. Therefore X is strictly convex.

Define

$$P(x_n) = (P_n x_n), \quad Q(x_n) = (Q_n x_n).$$

Since each P_n, Q_n is contractive on E_n , the operators P, Q are contractive for both $\|\cdot\|_{\infty}$ and G , hence for $\|\cdot\|_X$. They are non-zero projections, so $\|P\| = \|Q\| = 1$.

Let $T = PQ$ and let $x = (u_n)_{n=1}^{\infty} \in X$. Then

$$T^m x = ((-\rho_n^2)^m u_n)_{n=1}^{\infty}.$$

For each $m \geq 1$,

$$T^{2m} x - T^{2m+1} x = (\rho_n^{4m} (1 + \rho_n^2) u_n)_{n=1}^{\infty}.$$

Taking the supremum over n and using $\rho_n \uparrow 1$ gives

$$\|T^{2m} x - T^{2m+1} x\|_{\infty} = \sup_n \rho_n^{4m} (1 + \rho_n^2) = 2.$$

Thus

$$\|T^{2m} x - T^{2m+1} x\|_X \geq 2$$

for every m . The orbit $(T^m x)$ is not Cauchy, so $(PQ)^m$ does not converge strongly. $\square$

Complex variant

The proof above is written over the real field. If the source question is read as requiring complex-linear projections, the same construction should be run with the following balanced version of the two-dimensional block lemma: inside the closed bidisc of $\mathbb{C}^2$, take a balanced strictly convex body whose boundary contains the two circles

$$\{\lambda(1, -\rho) : |\lambda| = 1\}, \quad \{\lambda(\rho, 1) : |\lambda| = 1\},$$

and which is contained in the bidisc. The same rank-one formulas $P_{\rho} z = z_1(1, -\rho)$ and $Q_{\rho} z = z_2(\rho, 1)$ are then complex-linear contractive projections and the rest of the proof is unchanged. The real construction already answers the source paragraph as written; the balanced variant is included to flag the only extra point for a complex-field review.

Verification notes

The only geometric input is the elementary convex-body rounding in the two-dimensional block lemma. The rest of the construction is explicit. The space is not reflexive and not uniformly convex, as it contains an ℓ_{∞} -sum structure; this is consistent with the convergence theorem in the source paper.

A bounded novelty check was performed on 2026-06-14. The run indexes and prior archive notes were searched for arXiv:1006.2052, orthoprojections, strictly convex divergence, and averaged products of projections. Exact web searches for the source question and these keywords did not reveal a known answer. Novelty confidence is moderate pending human review.

Human-review recommendation

Check the planar convex-body block lemma carefully. If the intended scope is complex Banach spaces only, check the balanced complex variant stated above. Once that block lemma is accepted, the infinite-dimensional strictly convex sum and the divergence calculation are straightforward.

References

- [1] Catalin Badea and Yuri I. Lyubich, *Geometric, spectral and asymptotic properties of averaged products of projections in Banach spaces*, arXiv:1006.2052, 2010.

Human Review: Divergent orthoprojections in a strictly convex Banach space

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed construction appears to give a valid answer to the question in Badea-Lyubich asking whether divergent iterates of a product of orthoprojections can occur in a strictly convex Banach space.

The idea is to start from the elementary two-dimensional obstruction and then blow it up to infinite dimensions. For each $0 < \rho < 1$, the proof builds a two-dimensional strictly convex block E_ρ with two norm-one projections P_ρ, Q_ρ such that $P_\rho Q_\rho$ has eigenvalue $-\rho^2$. Taking $\rho_n \uparrow 1$ gives a sequence of blocks whose products behave more and more like multiplication by -1 .

Strict Convexity of the Sum

The key point in the infinite-dimensional construction is the renorming. One starts with the ℓ_∞ -sum

$$Y = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell_\infty},$$

and defines

$$\|x\|_X^2 = \|x\|_\infty^2 + \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{E_n}^2.$$

This norm is equivalent to the original ℓ_∞ -sum norm, since

$$\|x\|_\infty \leq \|x\|_X \leq \sqrt{2} \|x\|_\infty.$$

The strict convexity comes from the weighted ℓ_2 -term. Since each block E_n is strictly convex, the function

$$x_n \mapsto \|x_n\|_{E_n}^2$$

is strictly convex on E_n . Hence

$$x \mapsto \sum_{n=1}^{\infty} 2^{-n} \|x_n\|_{E_n}^2$$

is strictly convex on Y : if $x \neq y$, then $x_k \neq y_k$ for some coordinate k , and the positive weight 2^{-k} preserves the strict inequality coming from that block. The term $x \mapsto \|x\|_\infty^2$ is convex, so their sum $\|x\|_X^2$ is strictly convex. Therefore the norm $\|\cdot\|_X$ is strictly convex.

This also explains why the construction is not uniformly convex. The weights 2^{-n} tend to zero, so the added curvature becomes arbitrarily weak in far-out coordinates. Thus the renorming gives strict convexity without destroying the asymptotic ℓ_∞ -sum behaviour needed for the counterexample.

Contractivity and Divergence

The blockwise projections

$$P(x_n) = (P_n x_n), \quad Q(x_n) = (Q_n x_n)$$

remain contractive for the new norm. Indeed, they are contractive on each block, hence contractive for both the ℓ_∞ -term and the weighted ℓ_2 -term. Therefore they are norm-one projections for $\|\cdot\|_X$.

The divergence calculation is then straightforward. If u_n is the distinguished eigenvector in the n -th block and $x = (u_n)_n$, then for $T = PQ$,

$$T^m x = ((-\rho_n^2)^m u_n)_n.$$

Since $\rho_n \uparrow 1$, for every fixed m the far-out coordinates make $\rho_n^{4m}(1 + \rho_n^2)$ arbitrarily close to 2. Thus

$$\|T^{2m} x - T^{2m+1} x\|_\infty = 2,$$

and hence

$$\|T^{2m} x - T^{2m+1} x\|_X \geq 2.$$

So the orbit is not Cauchy, and the iterates $(PQ)^m$ do not converge strongly.

Review Outcome

The proof appears correct. The essential mechanism is clear: use two-dimensional almost- ℓ_∞^2 blocks with product eigenvalues $-\rho_n^2$, take $\rho_n \rightarrow 1$, and place the blocks in an ℓ_∞ -sum equipped with a small weighted ℓ_2 -correction. The correction is enough to force strict convexity, but not enough to remove the far-out oscillations responsible for divergence.

One minor point to keep in mind is the scalar field. The proof as written is most direct over the real field. If the intended reading of the source question is complex-linear, the balanced complex variant of the two-dimensional block should be checked explicitly. Apart from this standard-field issue, the argument appears sound.